

REIMS CHAMPAGNE, France

WMO: 070700

Lat: 49.310N

Lon: 4.050E

Elev: 95

StdP: 100.19

Time Zone: 1.00 (EUC)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.9	-5.3	-10.6	1.5	-6.3	-8.2	1.9	-4.4	13.3	9.6	11.9	9.9	1.9	30	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	30.6	20.5	28.5	19.8	26.6	19.2	21.6	28.4	20.7	26.9	19.7	25.2	3.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.1	24.0	18.5	13.5	23.2	17.7	12.8	22.3	63.2	28.4	59.8	26.9	56.7	25.3	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.1	8.0	-11.1	34.0	2.8	2.3	-13.1	35.6	-14.8	36.9	-16.4	38.2	-18.4	39.8		
(5)			-11.4	23.5	2.8	0.9	-13.5	24.1	-15.1	24.6	-16.7	25.1	-18.7	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	3.3	4.3	7.1	9.8	13.8	16.6	18.7	15.1	11.5	6.9	3.7	(6)	
(7)		DBStd	6.60	4.56	4.18	3.44	3.38	3.13	3.19	3.00	2.94	2.91	3.62	3.84	4.39	(7)
(8)		HDD10.0	854	209	163	101	43	6	0	0	0	1	27	105	199	(8)
(9)		HDD18.3	2851	465	394	348	254	143	71	31	31	104	213	342	454	(9)
(10)		CDD10.0	1162	3	2	11	38	124	199	274	269	153	72	13	3	(10)
(11)		CDD18.3	117	0	0	0	0	3	19	47	42	7	1	0	0	(11)
(12)		CDH23.3	1376	0	0	0	9	61	238	521	469	75	4	0	0	(12)
(13)		CDH26.7	421	0	0	0	0	7	61	170	172	12	0	0	0	(13)
(14)	Wind	WSAvg	3.7	4.3	4.2	4.1	3.8	3.5	3.2	3.3	3.1	3.3	3.7	3.9	4.2	(14)
(15)	Precipitation	PrecAvg	682	56	48	46	46	58	58	65	68	50	60	57	71	(15)
(16)		PrecMax	938	150	98	149	136	119	131	160	156	114	124	103	161	(16)
(17)		PrecMin	525	5	10	8	3	11	17	15	9	9	23	12	33	(17)
(18)		PrecStd	95	29	24	30	30	27	33	32	35	29	24	27	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	14.7	20.1	24.6	27.5	31.1	33.1	35.0	28.3	23.2	16.9	13.9	(19)
(20)			MCWB	11.7	10.4	12.6	16.1	19.8	21.9	20.8	21.2	19.6	17.9	14.2	12.1	(20)
(21)		2%	DB	12.4	12.7	17.2	21.9	24.9	28.3	30.7	30.6	25.5	20.4	15.0	12.5	(21)
(22)			MCWB	10.8	9.9	11.9	14.4	17.7	20.3	20.3	20.4	18.6	16.2	12.8	11.0	(22)
(23)		5%	DB	10.9	11.2	15.0	19.0	23.0	26.1	28.5	28.2	23.1	18.6	13.7	11.4	(23)
(24)			MCWB	9.4	9.2	10.9	13.2	16.7	19.4	19.7	19.5	17.6	15.4	11.9	10.1	(24)
(25)		10%	DB	9.6	10.1	13.1	16.7	20.9	23.9	26.3	25.8	20.9	17.0	12.4	10.0	(25)
(26)			MCWB	8.3	8.5	10.3	12.1	15.6	18.1	19.0	18.7	16.1	14.5	11.0	9.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	11.5	13.7	17.5	20.9	22.6	22.6	22.4	20.7	18.4	14.6	12.5	(27)
(28)			MCDWB	13.4	13.5	17.5	23.4	25.9	28.6	30.0	30.7	26.1	21.8	16.5	13.5	(28)
(29)		2%	WB	11.0	10.4	12.6	15.1	18.8	21.1	21.4	21.3	19.2	16.9	13.1	11.3	(29)
(30)			MCDWB	12.2	12.0	15.8	20.0	23.8	26.9	28.4	28.7	24.2	19.6	14.6	12.4	(30)
(31)		5%	WB	9.6	9.6	11.6	13.7	17.2	19.9	20.5	20.3	18.0	15.8	12.1	10.2	(31)
(32)			MCDWB	10.8	11.0	14.6	18.0	21.4	24.9	26.9	26.5	21.9	18.0	13.3	11.2	(32)
(33)		10%	WB	8.2	8.6	10.4	12.4	16.0	18.7	19.5	19.3	16.9	14.7	11.1	9.0	(33)
(34)			MCDWB	9.4	10.0	12.8	16.0	20.2	23.1	25.0	24.3	20.1	16.7	12.4	10.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.5	6.9	9.1	10.5	10.6	11.2	11.3	11.2	10.4	8.8	6.1	5.0	(35)
(36)			MCDBR	5.7	8.2	13.1	15.2	15.1	15.4	15.6	16.4	14.1	10.5	6.5	5.3	(36)
(37)			MCWBR	4.9	6.1	8.4	8.9	8.8	8.5	6.8	7.4	7.8	6.9	5.0	4.5	(37)
(38)		5% WB	MCDWB	5.6	7.0	10.4	13.1	13.2	13.7	13.7	14.0	11.8	9.6	5.9	5.1	(38)
(39)			MCWBR	5.0	5.6	7.0	7.9	8.2	7.9	6.6	6.6	6.9	6.6	4.8	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.342	0.375	0.404	0.413	0.414	0.412	0.404	0.374	0.362	0.333	0.316	(40)	
(41)		taud	2.416	2.369	2.276	2.209	2.221	2.252	2.251	2.300	2.364	2.396	2.414	2.386	(41)	
(42)		Ebn at Noon	731	798	827	840	847	847	842	832	821	763	703	675	(42)	
(43)		Edh at Noon	65	85	110	131	135	132	130	118	100	82	65	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.85	1.60	2.79	4.30	5.07	5.63	5.35	4.60	3.54	2.06	1.00	0.69	(44)	
(45)		RadStd	0.12	0.27	0.37	0.52	0.55	0.52	0.50	0.49	0.33	0.17	0.10	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page